

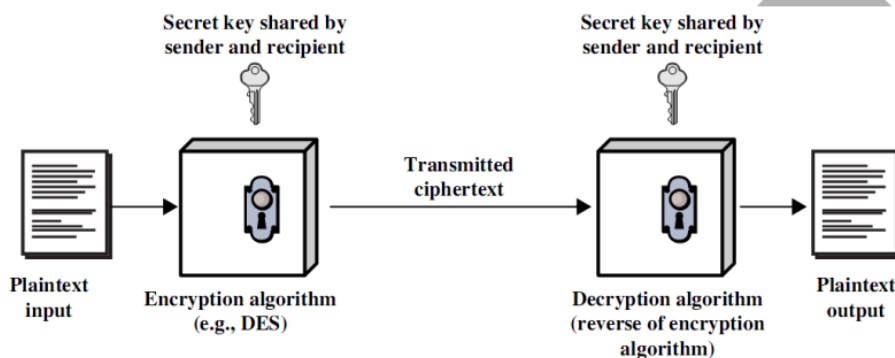
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Seventh Semester B.E. Degree Examination
CRYPTOGRAPHY

TIME: 03 Hours**Max. Marks: 100****00**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module-1 Download		COs	Marks
Q.01	a Define Cryptography & explain symmetric cipher model. Symmetric Cipher Model:  Cryptography is the study of techniques for securing communication and information by transforming it into a format that is unreadable to unauthorized users. This transformation process involves converting plaintext, which is the original intelligible message, into ciphertext, which is the encoded message. The reverse process, where ciphertext is converted back into plaintext, is known as decryption. The field encompasses various schemes and methods, collectively referred to as cryptographic systems or ciphers. the symmetric cipher model, which is a key aspect of cryptography. A symmetric encryption scheme consists of five essential components: 1. Plaintext : This is the original message or data that is input into the encryption algorithm. 2. Encryption Algorithm : This algorithm performs a series of substitutions and transformations on the plaintext to produce	1	7

	ciphertext. 3. Secret Key : A crucial element in symmetric encryption, the secret key is a value that is independent of both the plaintext and the algorithm. The specific key used will determine the output of the encryption process, meaning that different keys will yield different ciphertexts for the same plaintext. 4. Ciphertext : The output of the encryption process, this is the scrambled message that appears random and unintelligible without the corresponding key. 5. Decryption Algorithm : Essentially the reverse of the encryption algorithm, this takes the ciphertext and the secret key to produce the original plaintext. In symmetric encryption, both the sender and receiver share the same secret key, which must be kept secure. If an unauthorized party discovers the key, they can decrypt all messages encrypted with that key. This model was the only form of encryption used before the advent of public-key encryption in the 1970s and remains widely utilized today due to its efficiency and effectiveness. symmetric encryption relies on the secrecy of the key rather than the secrecy of the algorithm itself, allowing for broader implementation and use in various applications.		
b	Explain Ceaser Cipher with example The Caesar cipher is one of the simplest and earliest known encryption techniques, named after Julius Caesar, who reportedly used it to communicate with his generals. It is a type of substitution cipher where each letter in the plaintext is shifted a certain number of places down or up the alphabet. The most common shift used in the Caesar cipher is three, but any number from 1 to 25 can be used. How It Works: 1. Substitution : Each letter in the plaintext is replaced by a letter a fixed number of positions down the alphabet. For example, with a shift of 3: - A becomes D - B becomes E - C becomes F - ... - X becomes A - Y becomes B - Z becomes C 2. Encryption : To encrypt a message, you simply replace each letter with its corresponding letter from the shifted alphabet.	1	6

Example:

Let's take the plaintext message: "meet me after the toga party" and encrypt it using a Caesar cipher with a shift of 3.

- Plaintext : m e e t m e a f t e r t h e t o g a p a r t y
 - Ciphertext : P H H W P H D I W H U W K H W R J D
 S D U W B

So, the encrypted message becomes: "PHHW PH DIWHU WKH WRJD SDUWB" .

Decryption:

To decrypt the message, you simply reverse the process by shifting the letters back by the same number. For example, if you receive the ciphertext "PHHW PH DIWHU WKH WRJD SDUWB" , you would shift each letter back by 3 to retrieve the original plaintext.

The Caesar cipher is easy to understand and implement, but it is not very secure by modern standards, as there are only 25 possible keys (shifts), making it vulnerable to brute-force attacks. Nonetheless, it serves as a great introduction to the concepts of encryption and decryption!

1. Caesar Cipher:

The Caesar cipher involves replacing each letter of the alphabet with the letter standing three places further down the alphabet. For example,

plain: meet me after the toga party

cipher: PHHW PH DIWHU WKH WRJD SDUWB

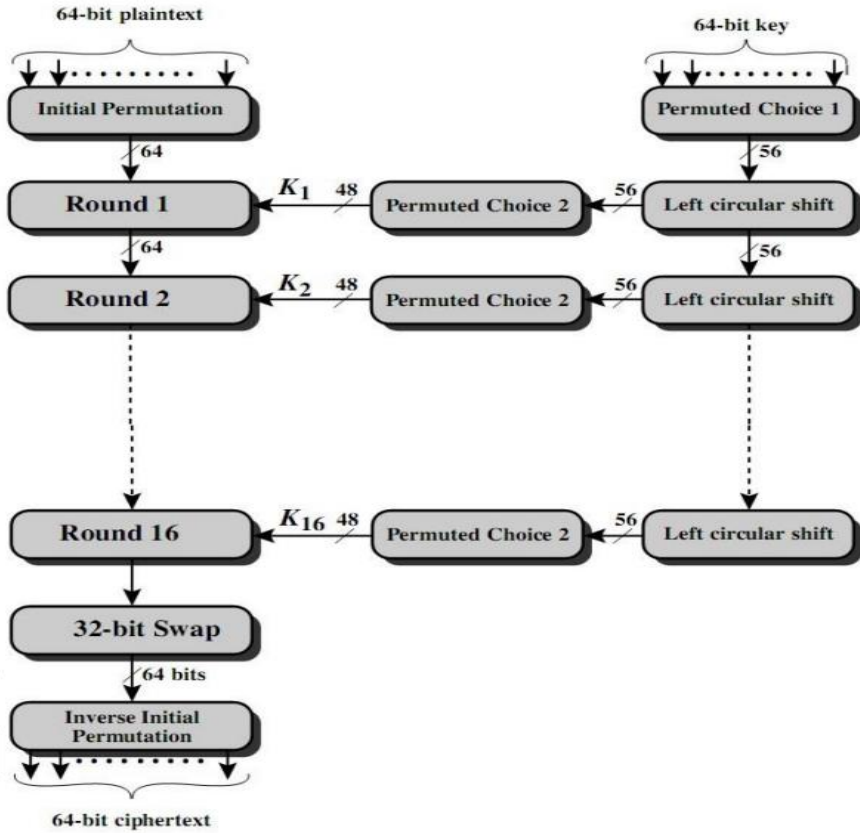
$$C = E(3, p) = (p + 3) \bmod 26$$

$$C = E(k, p) = (p + k) \bmod 26, \quad p = D(k, C) \bmod 26$$

Three important characteristics of this problem enabled us to use a brute-force cryptanalysis:

1. The encryption and decryption algorithms are known.
2. There are only 25 keys to try.
3. The language of the plaintext is known and easily recognizable.

Below Figure shows the results of applying this strategy to the example ciphertext. In this case, the plaintext leaps out as occupying the third line.

	c Explain Hill Cipher & apply it to encrypt Plain text = paymoremoney with key $K = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 9 \end{pmatrix}$	1	7
OR			
Q.0 2	a Explain working of DES Algorithm with Example The Data Encryption Standard (DES) is a symmetric encryption algorithm that was widely used for securing data until the introduction of the Advanced Encryption Standard (AES) in 2001. DES operates on 64-bit blocks of data using a 56-bit key, and it employs a series of transformations to encrypt the plaintext into ciphertext. 	1	10

Looking at the left-hand side of the figure, the processing of the plaintext proceeds in three phases.

1. The 64-bit plaintext passes through an **initial permutation (IP)** that rearranges the bits to produce the permuted input. This is followed by a phase consisting of **16 rounds** of the same function, which involves both permutation and substitution functions.
2. The output of the last (sixteenth) round consists of 64 bits that are a function of the input plaintext and the key. The left and right halves of the output are **swapped** to produce the *preoutput*.
3. Finally, the preoutput is passed through a permutation (**IP⁻¹**) that is the inverse of the initial permutation function, to produce the 64-bit ciphertext. With the exception of the initial and final permutations, DES has the exact structure of a Feistel cipher.

The right-hand portion shows the way in which the 56-bit key is used. Initially, the key is passed through a permutation function. Then, for each of the 16 rounds, a **subkey** (K_i) is produced by the combination of a left circular shift and a permutation. The permutation function is the same for each round, but a different subkey is produced because of the repeated shifts of the key bits.

Working of the DES Algorithm

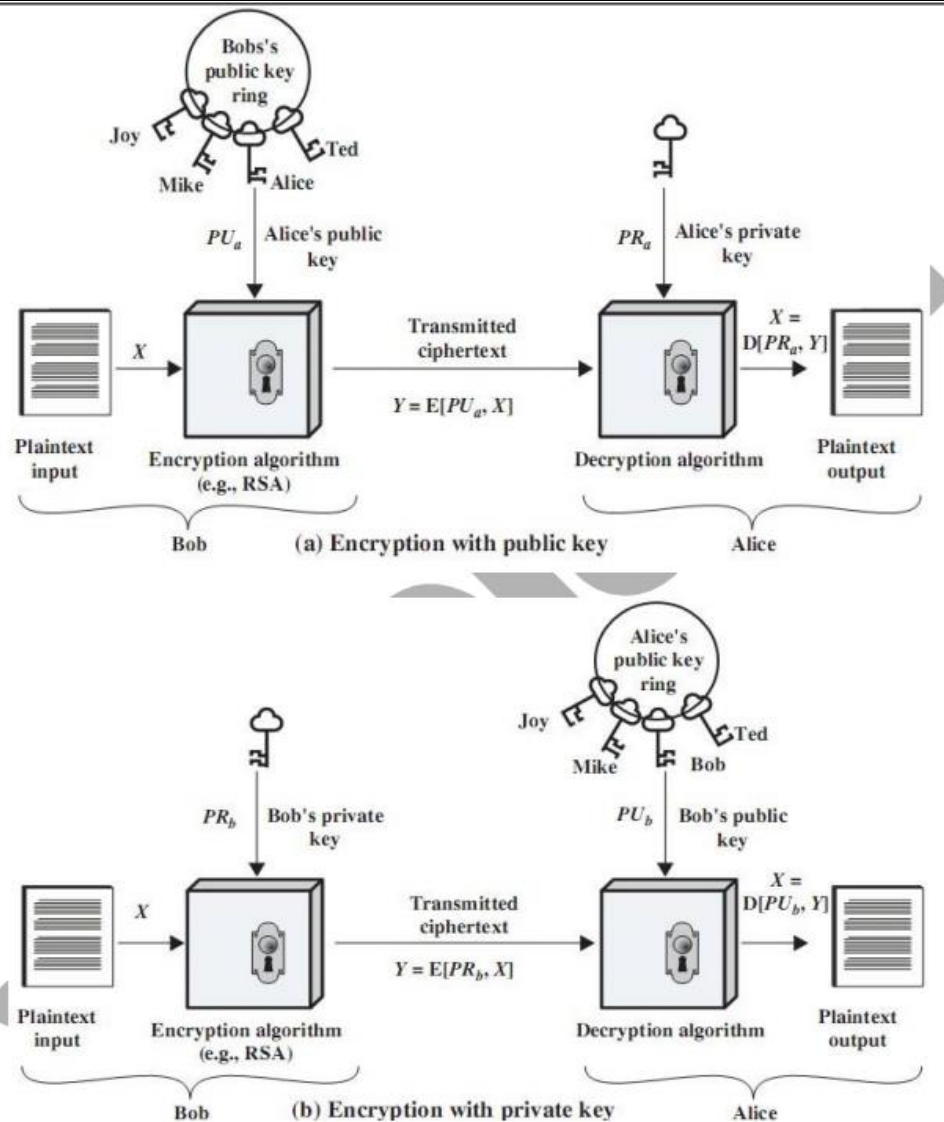
1. **Initial Permutation (IP)** : The 64-bit plaintext undergoes an initial permutation, which rearranges the bits according to a predefined table.
2. **Splitting the Data** : The permuted data is split into two halves: the left half (L) and the right half (R), each consisting of 32 bits.
3. **16 Rounds of Processing** : DES consists of 16 rounds of processing, where in each round, the following steps occur:
 - **Key Generation** : A 56-bit key is used to generate 16 subkeys ($K_1, K_2, \dots, K_{16}$) through a series of permutations and shifts.
 - **Function F** : The right half (R) is processed through a function F, which takes the right half and the subkey for that round. This function includes expansion, substitution using S-boxes, and permutation.
 - **XOR Operation** : The output of function F is XORed with the left half (L).
 - **Swapping** : After processing, the left and right halves are swapped.
4. **Final Permutation (IP⁻¹)** : After completing all 16 rounds, the final output is passed through the inverse of the initial permutation to produce the 64-bit ciphertext.

Example of DES Encryption

Let's consider an example to illustrate the DES encryption process:

- Plaintext : `02468aceeca86420` (in hexadecimal)

		- Key : `0f1571c947d9e859` (in hexadecimal) 1. Initial Permutation : The plaintext undergoes an initial permutation, resulting in a rearranged 64-bit block. 2. Splitting : The permuted block is split into L and R. 3. Rounds : For each of the 16 rounds: - Generate the subkey for that round. - Apply the function F to the right half and the subkey. - XOR the result with the left half and swap the halves. 4. Final Permutation : After 16 rounds, the final output is permuted to produce the ciphertext. The DES algorithm is a Feistel cipher, meaning that it uses the same algorithm for both encryption and decryption, with the order of the subkeys reversed during decryption. Although DES was once the standard for symmetric encryption, its key length of 56 bits is now considered insecure against modern brute-force attacks, leading to the adoption of more secure algorithms like AES and triple DES.		
	b	Explain Play Fair Cipher & Apply its rules for the following Key = MONARCHY Plaintext = CRYPTOGRAPHY to get the cipher text.	1	10
Module-2 Download				
Q. 03	a	Explain Public key Encryption And Decryption with example. Public key encryption and decryption is a method of securing communication that uses a pair of keys: a public key and a private key. The public key is shared openly, while the private key is kept secret by the owner. This system allows for secure data transmission without the need to share a secret key beforehand,	2	5



which is a significant advantage over traditional symmetric encryption.

How It Works:

- 1. Key Generation :** Each user generates a pair of keys. For example, let's say Alice and Bob are two users. Alice generates a public key (PU_a) and a private key (PR_a), while Bob generates his own public key (PU_b) and private key (PR_b).
- 2. Public Key Distribution :** Alice shares her public key (PU_a) with Bob, and Bob shares his public key (PU_b) with Alice. These public keys can be stored in a public register or shared through any accessible means.
- 3. Encryption :** When Bob wants to send a confidential message to Alice, he uses Alice's public key (PU_a) to encrypt the message. The encryption process can be represented as:

$$Y = E(PU_a, X)$$

where (X) is the plaintext message and (Y) is the resulting ciphertext.

4. Transmission : Bob sends the encrypted message (ciphertext (Y)) to Alice.

5. Decryption : Upon receiving the ciphertext, Alice uses her private key (PRa) to decrypt the message. The decryption process is represented as:

$$X = D(PR_a, Y)$$

This allows Alice to recover the original plaintext message (X 6666633).

Example:

Let's illustrate this with a simple example:

- Key Generation :

- Alice generates her keys: Public Key (PUa) = 12345, Private Key (PRa) = 67890.

- Bob generates his keys: Public Key (PUb) = 54321, Private Key (PRb) = 09876.

- Encryption :

- Bob wants to send the message "HELLO" to Alice. He encrypts it using Alice's public key (PUa):

$$Y = E(12345, \text{"HELLO"}) \quad \text{\text{(let's say this results in "XYZ")}}$$

- Transmission :

- Bob sends "XYZ" to Alice.

- Decryption :

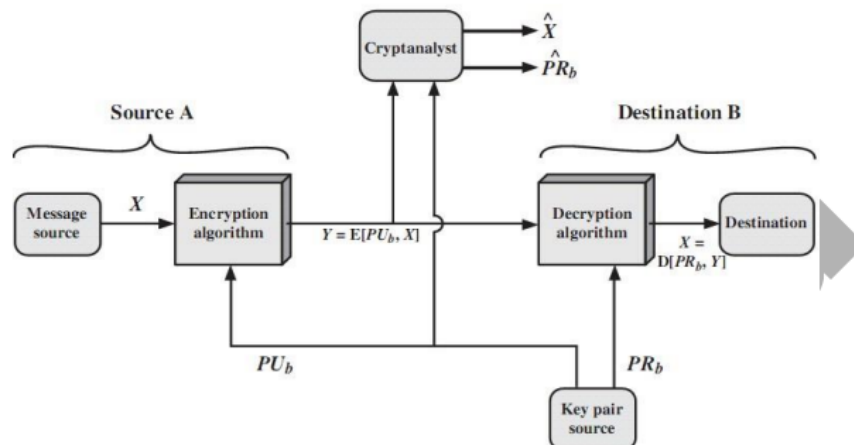
- Alice receives "XYZ" and decrypts it using her private key (PRa):

$$X = D(67890, \text{"XYZ"}) \quad \text{\text{(which results in "HELLO")}}$$

In this example, even if someone intercepts the message "XYZ", they cannot decrypt it without Alice's private key (PRa). This ensures the confidentiality of the communication. Public key encryption not only

		provides confidentiality but also supports digital signatures and secure key exchanges, making it a versatile tool in modern cryptography.		
	b	Explain the Description of RSA Algorithm .Assume 2prime no as p=3 and q=11 encryption key e =3 apply RSA algorithm & find cipher text for the plaintext m=8 & decrypt the cipher text to obtain plain text	2	10
	c	Explain elliptic curve cryptography algorithm Elliptic Curve Cryptography (ECC) is a form of public-key cryptography that utilizes the mathematics of elliptic curves to provide secure communication. The key advantage of ECC over traditional methods like RSA is that it offers equivalent security with much smaller key sizes, which reduces processing overhead and enhances efficiency, especially in environments with limited computational resources. Key Concepts of ECC: 1. Elliptic Curves : An elliptic curve is defined by a cubic equation in two variables, typically expressed in the form $(y^2 = x^3 + ax + b)$. For cryptographic purposes, the variables and coefficients are restricted to elements in a finite field, resulting in a finite abelian group. 2. Points on the Curve : The points on the elliptic curve, along with a special point at infinity, form a group under a defined addition operation. This addition operation is analogous to modular multiplication in RSA. 3. Private and Public Keys : - Each user selects a private key, which is a random integer. - The public key is generated by multiplying the private key with a predefined base point (G) on the elliptic curve, resulting in a point $(P_A = n_A \cdot G)$. 4. Key Exchange : The Diffie-Hellman key exchange can be implemented using ECC: - User A selects a private key (n_A) and computes their public key (P_A). - User B does the same with their private key (n_B) and public key (P_B). - They can then compute a shared secret key (k) using their private keys and the other user's public key: - $(k = n_A \cdot P_B = n_B \cdot P_A)$. 5. Encryption and Decryption :	2	5

		- To encrypt a message, the sender encodes the plaintext message as a point (P_m) on the elliptic curve. - A random integer (k) is chosen, and the ciphertext consists of two points: $(C_m = \{kG, P_m + kP_B\})$, where (P_B) is the recipient's public key. - The recipient can decrypt the message by using their private key to recover the original point (P_m). Security Basis: The security of ECC relies on the difficulty of the discrete logarithm problem in the context of elliptic curves. While it is easy to compute $(Q = kP)$ given (k) and (P), it is computationally hard to determine (k) given (Q) and (P). This property makes ECC a robust choice for secure communications. Applications: ECC is widely used in various applications, including secure web communications (SSL/TLS), digital signatures, and mobile devices, where efficiency and security are paramount. elliptic curve cryptography provides a powerful and efficient means of securing communications through the use of elliptic curves, making it a popular choice in modern cryptographic systems.		
OR				
Q.0 4	a	Explain Secrecy and Authentication in public key cryptosystem In a public-key cryptosystem, secrecy and authentication are achieved through the use of two distinct keys: a public key, which is shared openly, and a private key, which is kept secret by the owner.	2	8



Public-Key Cryptosystems: Secrecy and Authentication

- Public-key schemes can be used for either secrecy or authentication, or both (as shown here). There is some source A that produces a message in plaintext X. The message is intended for destination B.
- B generates a related pair of keys: a public key, PU_b , and a private key, PR_b . PR_b is known only to B, whereas PU_b is publicly available and therefore accessible by A.
- With the message X and the encryption key PU_b as input, A forms the ciphertext

$$Y = E(PU_b, X)$$

- The intended receiver, in possession of the matching private key, is able to invert the transformation:
 $X = D(PR_b, Y)$.
- An adversary, observing Y and having access to PU_b , but not having access to PR_b or X, must attempt to recover X and/or PR_b . This provides confidentiality.
- Public-key encryption can also provide authentication:

$$Y = E(PR_a, X); X = D(PU_a, Y)$$

- To provide both the authentication function and confidentiality have a double use of the public-key scheme (as shown here):

$$Z = E(PU_b, E(PR_a, X))$$

$$X = D(PU_a, D(PR_b, Z))$$

Secrecy is primarily ensured through the encryption process. When a sender wants to send a confidential message to a recipient, they encrypt the message using the recipient's public key. This means that only the recipient, who possesses the corresponding private key, can decrypt the message and read its contents. This method guarantees that even if the encrypted message is intercepted by a third party, they cannot decrypt it without access to the private key. However, it's important to note that while this process provides confidentiality, it does not inherently protect against eavesdropping unless additional measures are taken.

Authentication, on the other hand, is achieved through the use of digital signatures. In this process, the sender encrypts a message using their own private key. When the recipient receives the message, they can decrypt it using the sender's public key. Since only the sender has access to their private key, this confirms that the message was indeed sent by them, thus authenticating the source. Additionally, because the

message is encrypted with the sender's private key, any alteration to the message would require access to that private key, making it infeasible for an attacker to modify the message without detection.

secrecy in a public-key cryptosystem is achieved through the encryption of messages with the recipient's public key, ensuring that only the intended recipient can read them. Authentication is provided by the sender signing the message with their private key, allowing the recipient to verify the sender's identity and the integrity of the message. This dual approach enhances the security of communications in various contexts, making public-key cryptography a powerful tool for secure communication.

b Explain diffie Hellman key exchange algorithm with example.

2

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Diffie-Hellman Key Exchange

The purpose of the algorithm is to enable two users to securely exchange a key that can then be used for subsequent encryption of messages.

If a is a "primitive root of the prime number p , then the numbers

$$a \bmod p, a^2 \bmod p, \dots, a^{p-1} \bmod p$$

are distinct and consist of the integers from 1 through $p-1$ in some permutation.

Example:

3 is primitive root of 7.

The number 3 is a primitive root modulo 7, because

$$3^1 = 3 = 3^0 \times 3 \equiv 3 \pmod{7}$$

$$3^2 = 9 = 3^2 \times 3 \equiv 9 \equiv 2 \pmod{7}$$

$$3^3 = 27 = 3^3 \times 3 \equiv 6 \equiv 6 \pmod{7}$$

$$3^4 = 81 = 3^4 \times 3 \equiv 18 \equiv 4 \pmod{7}$$

$$3^5 = 243 = 3^5 \times 3 \equiv 12 \equiv 5 \pmod{7}$$

$$3^6 = 729 = 3^6 \times 3 \equiv 15 \equiv 1 \pmod{7}$$

Here we see that the period of 3^k modulo 7 is 6. The remainders in the period, which are 3, 2, 6, 4, 5, 1, form a rearrangement of all nonzero remainders modulo 7, implying that 3 is indeed a primitive root modulo 7.

The Algorithm:

There are two publicly known numbers: a prime number q and an integer that is a primitive root of q .

Suppose the users A and B wish to exchange a key. User A selects a random integer $X_A < q$ and computes $Y_A = \alpha^{X_A} \bmod q$. Similarly, user B independently selects a random integer $X_B < q$ and computes $Y_B = \alpha^{X_B} \bmod q$.

Each side keeps the X value private and makes the Y value available publicly to the otherside.

User A computes the key as $K = (Y_B)^X \mod q$ and user B computes the key as $K = (Y_A)^{X_B} \mod q$.

These two calculations produce identical results:

$$\begin{aligned}
 K &= (Y_B)^{X_A} \mod q \\
 &= (\alpha^{X_B} \mod q)^{X_A} \mod q \\
 &= (\alpha^{X_B})^{X_A} \mod q && \text{by the rules of modular arithmetic} \\
 &= \alpha^{X_B X_A} \mod q \\
 &= (\alpha^{X_A})^{X_B} \mod q \\
 &= (\alpha^{X_A} \mod q)^{X_B} \mod q \\
 &= (\alpha^{X_A} \mod q)^{X_B} \mod q \\
 &= (Y_A)^{X_B} \mod q
 \end{aligned}$$

Example:

Prime number $q = 353$ and a primitive root of 353, in this case $\alpha = 3$.

A and B select secret keys $X_A = 97$ and $X_B = 233$, respectively. Each computes its publickey:

A computes $Y_A = 3^{97} \mod 353 = 40$.

B computes $Y_B = 3^{233} \mod 353 = 248$.

After they exchange public keys, each can compute the common secret key:

A computes $K = (Y_B)^{X_A} \mod 353 = 248^{97} \mod 353 = 160$.

B computes $K = (Y_A)^{X_B} \mod 353 = 40^{233} \mod 353 = 160$.

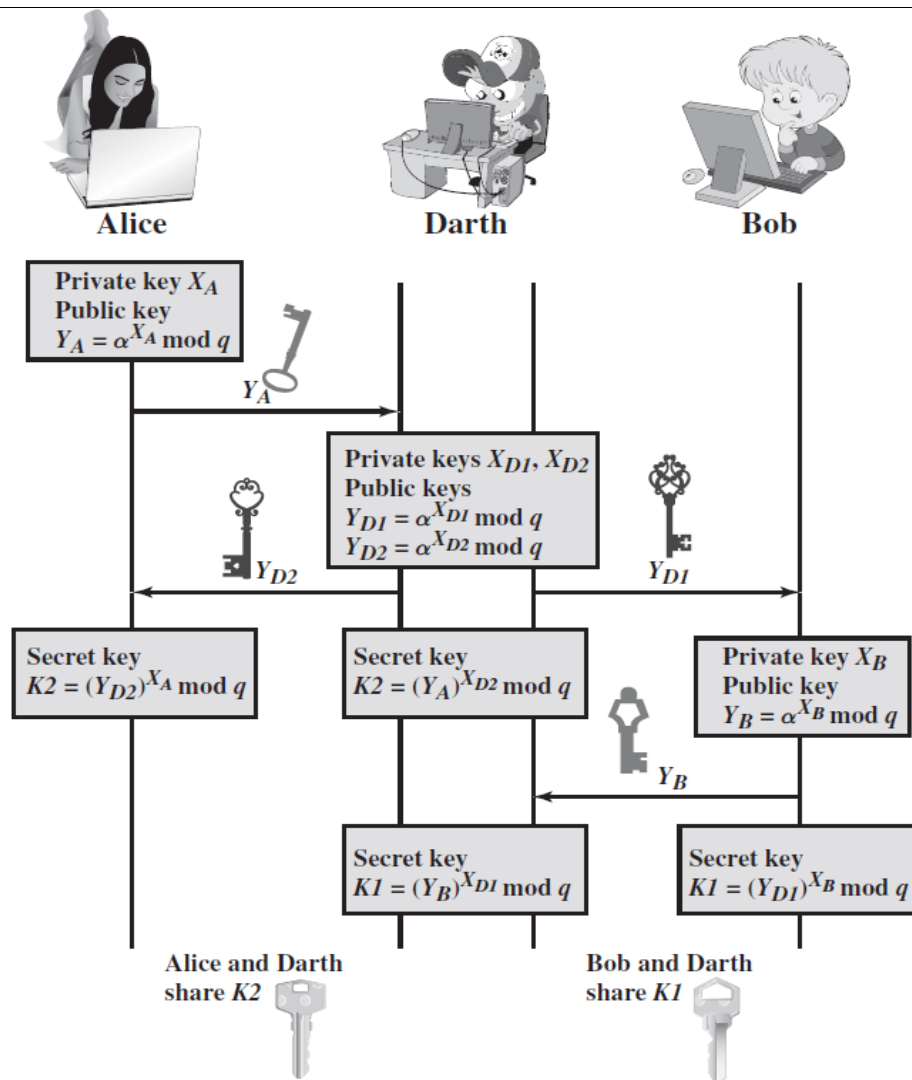
Key Exchange Protocols:

c Explain man in middle attack with example.

A man-in-the-middle (MitM) attack is a type of cyberattack where an adversary secretly intercepts and relays messages between two parties who believe they are communicating directly with each other. This attack can compromise the confidentiality and integrity of the communication.

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Example of a Man-in-the-Middle Attack:

Let's consider a scenario involving two users, Alice and Bob, who want to exchange keys securely. An attacker, whom we'll call Darth, aims to intercept their communication.

1. Preparation : Darth generates two random private keys, (X_{D1}) and (X_{D2}), and computes the corresponding public keys, (Y_{D1}) and (Y_{D2}).
2. Key Exchange Initiation : Alice sends her public key (Y_A) to Bob.
3. Interception : Instead of Bob receiving Alice's public key, Darth intercepts (Y_A) and sends his own public key (Y_{D1}) to Bob.
4. Key Calculation : Bob receives (Y_{D1}) and calculates a secret key ($K1$) using (Y_{D1}) and his private key (X_B):

$$K1 = (Y_{D1})^{X_B} \bmod q$$

5. Darth's Secret Key : Meanwhile, Darth calculates a different secret key (K2) using Alice's public key (YA) and his own private key (XD2):

$$K2 = (YA)^{XD2} \mod q$$

6. Communication : Now, both Darth and Bob think they are communicating securely with Alice, but in reality, Darth can read and modify the messages exchanged between them.

This attack highlights the importance of ensuring that public keys are verified and authenticated before use, as the lack of such verification allows an attacker to insert themselves into the communication channel undetected.

a MitM attack can lead to unauthorized access to sensitive information, and it is crucial to implement measures such as digital signatures or public-key certificates to prevent such vulnerabilities.

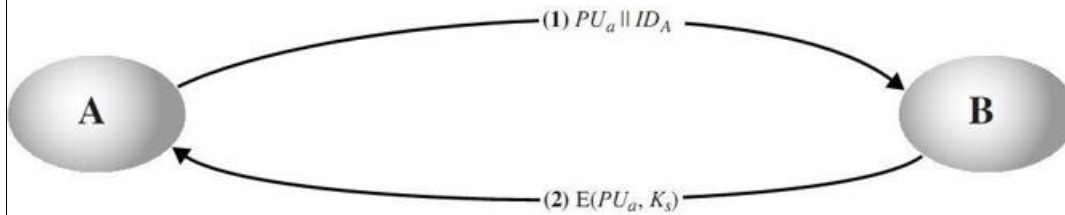
Module-3

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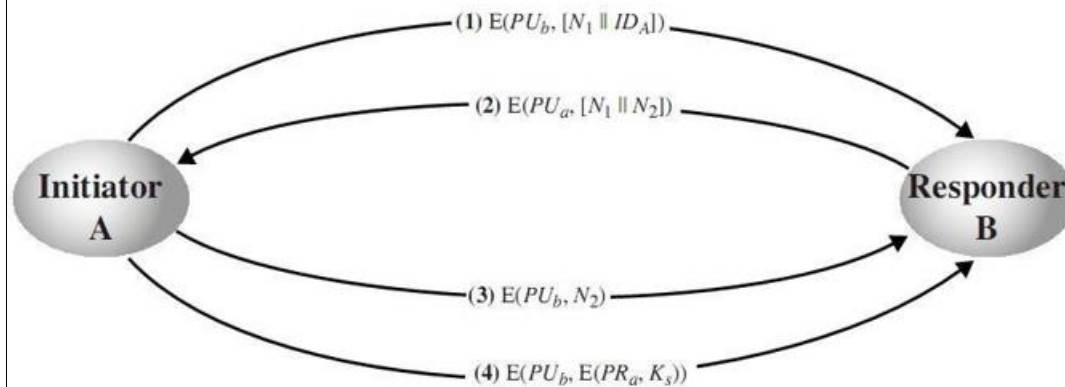
Q. 05	a	Explain symmetric key distribution using symmetric encryption Symmetric key distribution using symmetric encryption involves several methods for securely sharing a key between two parties, typically referred to as A and B. Here are the main approaches outlined in the document: 1. Physical Delivery : A can select a key and physically deliver it to B. This method is straightforward but can be impractical for remote communications. 2. Third-Party Delivery : A third party can select the key and physically deliver it to both A and B. This adds a layer of trust but still relies on physical transport. 3. Using an Old Key : If A and B have previously used a key, one party can transmit a new key to the other, encrypting it with the old key. This method is efficient but poses a risk if the old key is compromised. 4. Key Distribution Center (KDC) : If A and B each have an encrypted connection to a third party (the KDC), the KDC can deliver a key over	3	10
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	these secure links. This is a widely adopted method for end-to-end encryption, especially in larger networks. The document emphasizes that for symmetric encryption to be effective, the shared key must be protected from unauthorized access. Frequent key changes are recommended to limit the potential damage if a key is compromised. The strength of the cryptographic system heavily relies on the key distribution technique used. In a distributed system, the challenge of key distribution increases significantly, especially when many hosts or users need to communicate securely. For instance, if there are N entities that wish to communicate, the number of required session keys can be as high as $([N(N-1)]/2)$, while only N master keys are needed, which can be distributed more easily. symmetric key distribution is a critical aspect of secure communications, and understanding the various methods and their implications is essential for effective key management.		
b	Explain major issues with key distribution center (KDC). The key distribution center (KDC) plays a crucial role in symmetric key management, but it comes with several significant issues: 1. Trust Requirement : The KDC must be a trusted entity. If the KDC is compromised, an attacker could potentially distribute counterfeit keys, leading to unauthorized access and data breaches. This trust requirement can be a single point of failure in the security architecture. 2. Decentralization Challenges : While a single KDC may work for smaller networks, it becomes impractical for larger networks. To address this, a hierarchical structure of KDCs can be established, where local KDCs manage key distribution within smaller domains. However, this adds complexity to the system and requires secure communication between KDCs. 3. Scalability Issues : As the number of users and devices increases, the KDC can become a bottleneck. The automated key distribution approach is designed to alleviate this, but it still requires careful management to ensure that the KDC can handle the load without delays. 4. Session Key Management : The KDC is responsible for generating and distributing session keys. Frequent key changes are necessary to enhance security, but this can delay the start of communications and burden network capacity. A balance must be struck between security	3	10

		and performance. 5. Potential for Subversion : If a KDC is subverted, the damage can be extensive, especially if it serves a large network. A compromised KDC can lead to widespread vulnerabilities, as it controls the distribution of keys across multiple domains. 6. Public-Key Integration : While integrating public-key infrastructure can enhance security, it also introduces additional complexity and computational overhead. The hybrid approach, which combines symmetric and asymmetric encryption, can degrade overall system performance if not managed properly. 7. Key Hierarchy Management : Establishing a hierarchy of keys can minimize the effort involved in master key distribution, but it also requires careful management to ensure that each level of the hierarchy is secure and that the master keys are protected from unauthorized access. while KDCs are essential for secure key distribution in symmetric encryption, they introduce various challenges related to trust, scalability, management, and potential vulnerabilities that must be carefully addressed to maintain a secure cryptographic environment.		
OR				
Q. 06	a	Explain symmetric key distribution using asymmetric encryption Symmetric key distribution using asymmetric encryption is a method that leverages the strengths of both encryption types to securely share secret keys between parties. Here's how it works in detail:	3	10



Simple Secret Key Distribution



Secret Key Distribution with Confidentiality and Authentication

1. Key Generation : Each party, say Alice and Bob, generates a pair of keys: a public key and a private key. Alice's keys are denoted as (PU_A) (public key) and (PR_A) (private key), while Bob's keys are (PU_B) and (PR_B).

2. Public Key Exchange : Alice sends her public key (PU_A) to Bob along with her identifier (ID_A). This allows Bob to know who he is communicating with and to use Alice's public key for secure communication.

3. Secret Key Creation : Bob generates a secret symmetric key (K_s) that they will use for their communication. This key is crucial because symmetric encryption requires both parties to share the same key.

4. Encryption of the Secret Key : Bob encrypts the secret key (K_s) using Alice's public key (PU_A). This ensures that only Alice, who possesses the corresponding private key (PR_A), can decrypt this message and retrieve the secret key.

5. Sending the Encrypted Key : Bob sends the encrypted secret key to Alice. Since it is encrypted with Alice's public key, it cannot be read by anyone else, ensuring confidentiality.

6. Decryption by Alice : Upon receiving the encrypted secret key, Alice uses her private key (PR_A) to decrypt the message and obtain the symmetric key (K_s).

7. Secure Communication : Now that both Alice and Bob have the same symmetric key (K_s), they can use it for secure communication using symmetric encryption methods.

This method is advantageous because it combines the efficiency of symmetric encryption for data transmission with the security of asymmetric encryption for key distribution. However, it is important to note that while public-key systems are not typically used for encrypting large amounts of data due to their inefficiency, they are ideal for securely transmitting small pieces of information, such as secret keys.

asymmetric encryption facilitates the secure distribution of symmetric keys, allowing parties to communicate securely without the risk of key interception during the exchange process.

b Explain different ways of distributing Public keys

3

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There are several techniques for distributing public keys, each with its own advantages and vulnerabilities. Here are the main methods:

1. **Public Announcement** : This method allows participants to broadcast their public keys to the community. For example, users of PGP (Pretty Good Privacy) often append their public keys to



Figure 10.1 Uncontrolled Public Key Distribution

messages sent in public forums like USENET newsgroups. While convenient, this approach is susceptible to forgery, as anyone can impersonate another user and send a counterfeit public key.

2. Publicly Available Directory : In this scheme, a trusted entity maintains a dynamic directory of public keys. Each participant registers their public key with this authority, which ensures that the keys are

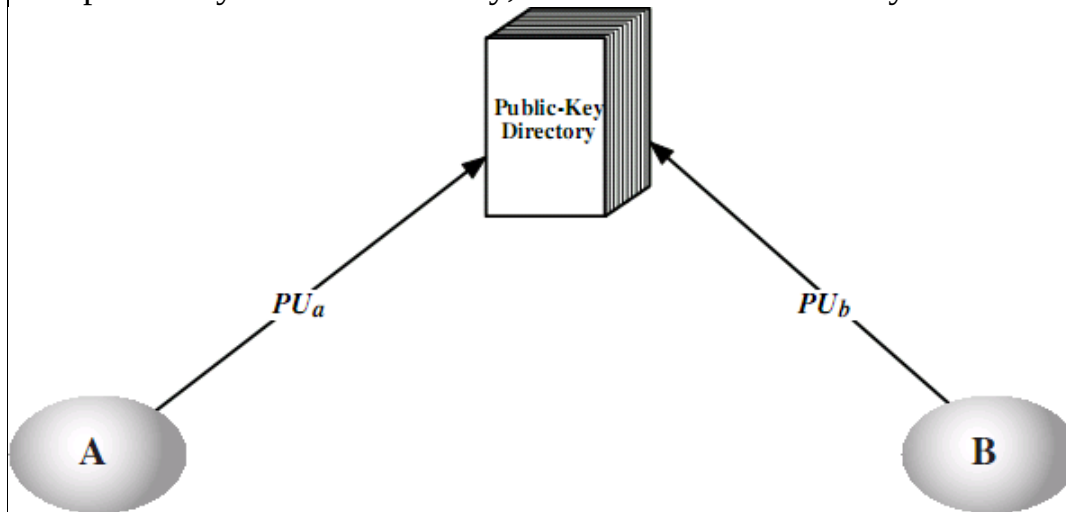
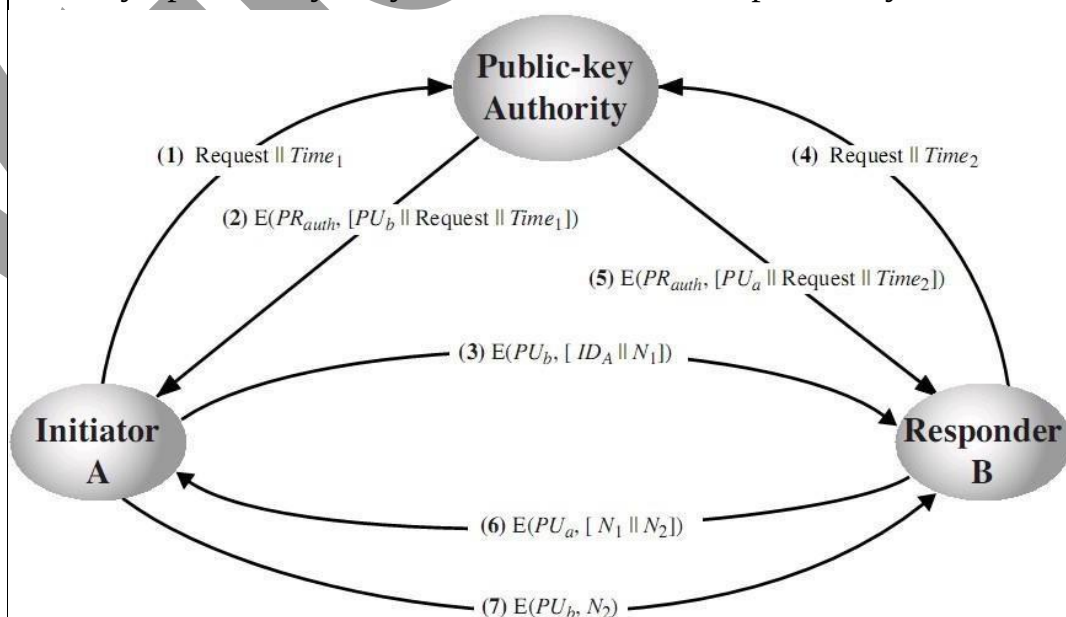


Figure 10.2 Public Key Publication

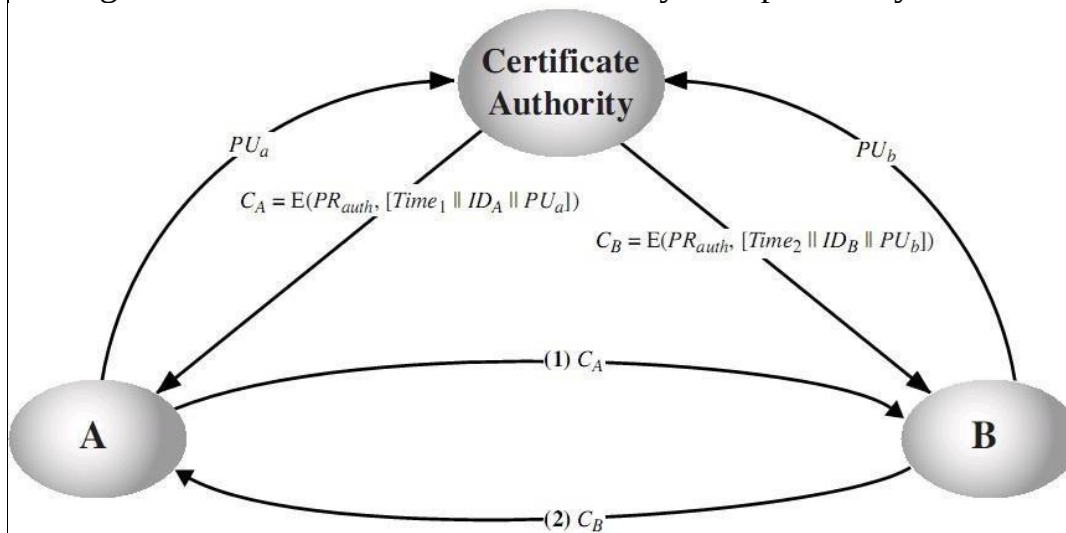
authentic. This method enhances security but requires secure registration processes, such as in-person verification or secure authenticated communication.

3. Public-Key Authority : This approach involves a central authority that controls the distribution of public keys. The authority maintains a directory of public keys and ensures that each participant's key is registered securely. However, if an adversary gains access to the authority's private key, they could issue counterfeit public keys.



4. Public-Key Certificates : Certificates are issued by a trusted third

party, known as a certificate authority (CA). A certificate contains a public key, the identity of the key owner, and a digital signature from the CA. This allows users to verify the authenticity of a public key without needing to contact the CA directly. Users can publish their certificates, making it easier for others to obtain and verify their public keys.



Each of these methods has its own strengths and weaknesses, and the choice of which to use often depends on the specific security requirements and context of the communication.

Module-4

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Q. 07 a Explain X.509 CERTIFICATE with format.

4

10

X.509 certificates are a crucial component of public-key infrastructure (PKI) and are widely used for secure communications over networks. They serve as a means of associating a public key with the identity of the certificate holder, typically a user or an organization. The X.509 standard, which is part of the X.500 series of recommendations, defines the structure and format of these certificates.

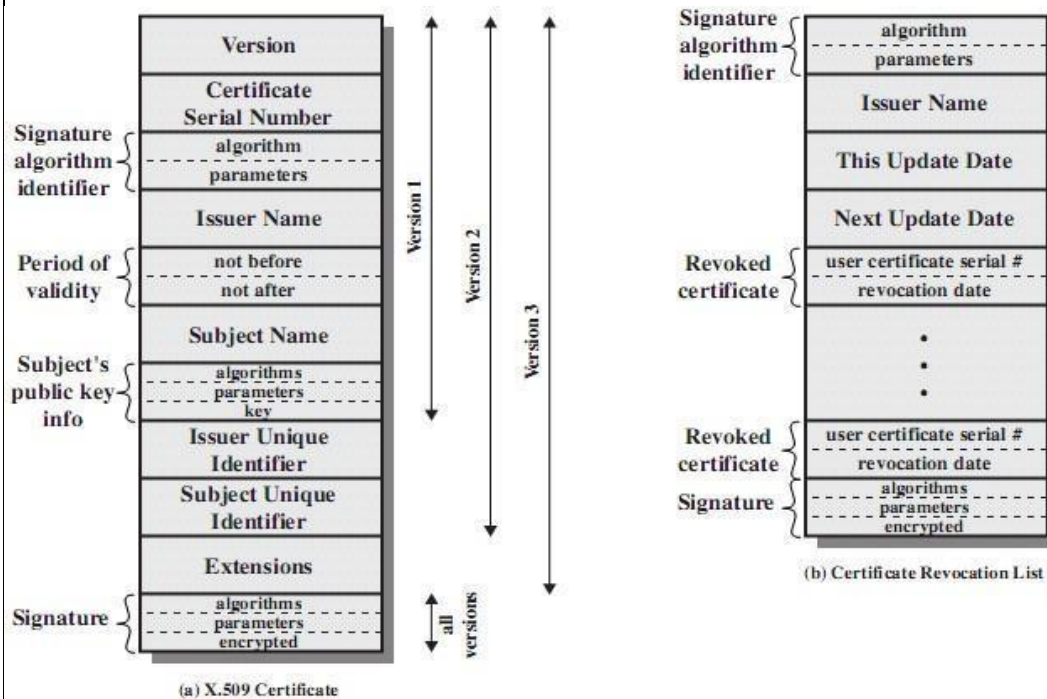


Figure 14.4 X.509 Formats

Format of an X.509 Certificate

An X.509 certificate generally includes the following elements:

1. **Version** : This indicates the version of the certificate format. The default is version 1. If the issuer unique identifier or subject unique identifier is present, it must be version 2. If one or more extensions are included, it must be version 3.
2. **Serial Number** : A unique integer value assigned by the issuing Certification Authority (CA) that distinguishes this certificate from others.
3. **Signature Algorithm Identifier** : This specifies the algorithm used to sign the certificate, along with any associated parameters. This information is also repeated in the signature field at the end of the certificate.
4. **Issuer Name** : The X.500 name of the CA that created and signed the certificate.
5. **Period of Validity** : This consists of two dates: the start date (Not Before) and the end date (Not After) during which the certificate is

	valid. 6. Subject Name : The name of the entity (user or organization) to which the certificate refers. This is the identity that the certificate certifies. 7. Subject's Public Key Information : This includes the public key of the subject, along with an identifier for the algorithm used and any associated parameters. 8. Issuer Unique Identifier : An optional field that uniquely identifies the issuing CA, particularly useful if the X.500 name has been reused. 9. Subject Unique Identifier : Another optional field that uniquely identifies the subject of the certificate. 10. Extensions : These are additional fields that can provide extra information about the certificate, such as usage constraints or policies. Importance of X.509 Certificates X.509 certificates are essential for establishing secure communications in various applications, including SSL/TLS for web security, S/MIME for email security, and IP security. They help ensure that the public keys used in these protocols are authentic and have not been tampered with, thereby enabling secure data exchange over potentially insecure networks. X.509 certificates play a vital role in the realm of digital security, providing a standardized way to verify identities and secure communications.		
b	Explain Remote user-authentication using symmetric encryption Remote user authentication using symmetric encryption involves a series of steps that ensure the identity of users trying to access a network or system. The process typically relies on a trusted Key Distribution Center (KDC) to facilitate secure communication between users and services. Here's a detailed breakdown of how it works: 1. Initial Request : When a user (let's call them User A) wants to authenticate to a service (Server B), they first send a request to the KDC. This request includes User A's ID and the ID of Server B, along with a nonce (a random number used once) to ensure the request is fresh and to prevent replay attacks.	4	10

		2. KDC Verification : The KDC checks its database to verify that User A is legitimate and has the correct credentials. If the credentials are valid, the KDC generates a session key (K_s) that will be used for the communication session between User A and Server B. 3. Ticket Generation : The KDC creates a ticket that contains User A's ID, the network address of Server B, and the session key (K_s). This ticket is encrypted using a secret key shared between the KDC and Server B, ensuring that only Server B can decrypt it. 4. Sending the Ticket : The KDC sends the encrypted ticket back to User A. Along with the ticket, the KDC also sends a message encrypted with a key shared between User A and the KDC, which includes the session key (K_s) and the nonce that User A originally sent. 5. Accessing the Server : User A then forwards the ticket to Server B along with the nonce. Server B decrypts the ticket using its secret key, retrieves the session key (K_s), and verifies that the nonce matches the one sent by User A. This confirms that the request is timely and not a replay. 6. Session Establishment : Once Server B verifies the ticket and the nonce, it can establish a secure session with User A using the session key (K_s). This key is used for encrypting the communication between User A and Server B, ensuring confidentiality and integrity. This method of authentication is efficient and secure, as it minimizes the exposure of sensitive information and relies on symmetric encryption to protect the data exchanged during the authentication process. Additionally, it addresses key issues such as confidentiality and timeliness, which are crucial in preventing unauthorized access and replay attacks.		
OR				
Q. 08	a	Explain Kerberos Authentication Service with version 4 Dialogue. Kerberos Authentication Service, particularly in version 4, is designed to facilitate secure authentication in a distributed network environment. It operates on the principle of mutual trust between clients and servers, relying on a centralized authentication server (AS) to mediate this trust.	4	10

Table 14.1 Summary of Kerberos Version 4 Message Exchanges

(1) $C \rightarrow AS \quad ID_C \parallel ID_{TGS} \parallel TS_1$ (2) $AS \rightarrow C \quad E(K_c, [K_{c,tgs} \parallel ID_{TGS} \parallel TS_2 \parallel Lifetime_2 \parallel Ticket_{TGS}])$ $Ticket_{TGS} = E(K_{TGS}, [K_{c,tgs} \parallel ID_C \parallel AD_C \parallel ID_{TGS} \parallel TS_2 \parallel Lifetime_2])$ (a) Authentication Service Exchange to obtain ticket-granting ticket
(3) $C \rightarrow TGS \quad ID_V \parallel Ticket_{TGS} \parallel Authenticator_c$ (4) $TGS \rightarrow C \quad E(K_{c,tgs}, [K_{c,v} \parallel ID_V \parallel TS_4 \parallel Ticket_v])$ $Ticket_{TGS} = E(K_{TGS}, [K_{c,tgs} \parallel ID_C \parallel AD_C \parallel ID_{TGS} \parallel TS_2 \parallel Lifetime_2])$ $Ticket_v = E(K_v, [K_{c,v} \parallel ID_C \parallel AD_C \parallel ID_V \parallel TS_4 \parallel Lifetime_4])$ $Authenticator_c = E(K_{c,tgs}, [ID_C \parallel AD_C \parallel TS_3])$ (b) Ticket-Granting Service Exchange to obtain service-granting ticket
(5) $C \rightarrow V \quad Ticket_v \parallel Authenticator_c$ (6) $V \rightarrow C \quad E(K_{c,v}, [TS_5 + 1])$ (for mutual authentication) $Ticket_v = E(K_v, [K_{c,v} \parallel ID_C \parallel AD_C \parallel ID_V \parallel TS_4 \parallel Lifetime_4])$ $Authenticator_c = E(K_{c,v}, [ID_C \parallel AD_C \parallel TS_5])$ (c) Client/Server Authentication Exchange to obtain service

In version 4, the authentication process involves several key steps, which can be illustrated through a simple dialogue:

1. Client Request for Ticket-Granting Ticket (TGT) : The client initiates the process by sending a request to the Authentication Server (AS) for a Ticket-Granting Ticket (TGT). This request includes the client's identity (ID_C) and the identity of the Ticket Granting Service (ID_{TGS}) that the client wishes to access.
2. AS Response with TGT : Upon receiving the request, the AS verifies the client's identity and, if valid, responds with a TGT. This ticket is encrypted using a key derived from the user's password (K_c), ensuring that only the client can decrypt it. The TGT contains a session key ($K_{c, tgs}$) for secure communication between the client and the TGS, along with timestamps and the ticket's lifetime.
3. Client Requests Service-Granting Ticket : The client then uses the TGT to request a service-granting ticket from the TGS. This request includes the TGT and an authenticator that contains the client's name, network address, and a timestamp.
4. TGS Issues Service-Granting Ticket : The TGS decrypts the TGT

	and authenticator, verifies the request, and issues a service-granting ticket for the requested application server. This ticket allows the client to access the desired service. 5. Client Accesses the Service : Finally, the client sends the service-granting ticket and another authenticator to the application server. The server verifies the ticket and authenticator, and if everything checks out, grants access to the requested service. This dialogue illustrates the essence of the Kerberos version 4 protocol, which, while effective, has limitations such as reliance on the DES encryption standard and lack of support for interrealm authentication. These shortcomings were addressed in version 5, which introduced enhancements like improved encryption flexibility and the ability to forward credentials across different realms.		
b	Explain Remote user-authentication using asymmetric encryption. Remote user authentication using asymmetric encryption involves a series of steps that leverage public and private key pairs to ensure secure communication and identity verification over a network. Here's a detailed breakdown of the process: 1. Key Pair Generation : Each user generates a pair of keys: a public key, which can be shared with anyone, and a private key, which must be kept secret. The public key is used for encryption, while the private key is used for decryption. 2. Authentication Request : When a user (let's call them User A) wants to authenticate to a server (Server B), User A sends a request to Server B. This request typically includes User A's identifier (ID_A) and a timestamp to prevent replay attacks. 3. Challenge from the Server : Upon receiving the request, Server B generates a challenge, which is a random number (nonce) to ensure the freshness of the authentication attempt. Server B encrypts this challenge using User A's public key and sends it back to User A. 4. Response from the User : User A receives the encrypted challenge and decrypts it using their private key. User A then responds by encrypting the challenge again, along with their own nonce (to prove they are the one responding) using Server B's public key, and sends this back to Server B. 5. Verification by the Server : Server B receives the response and decrypts it using its private key. If the decrypted challenge matches the	4	10

original challenge it sent, and the nonce from User A is valid, Server B can confirm User A's identity.

6. Session Key Establishment : Once authenticated, both parties can establish a session key for secure communication. This session key can be used for symmetric encryption during their interaction, ensuring confidentiality and integrity of the data exchanged.

This method of authentication is particularly robust because it does not require the transmission of sensitive information, such as passwords, over the network. Instead, it relies on the security of the private keys and the encryption process, making it resistant to eavesdropping and replay attacks. Additionally, the use of timestamps and nonces helps to ensure that each authentication attempt is unique and timely, further enhancing security.

asymmetric encryption provides a secure framework for remote user authentication, allowing users to verify their identities without exposing sensitive credentials.

Module-5

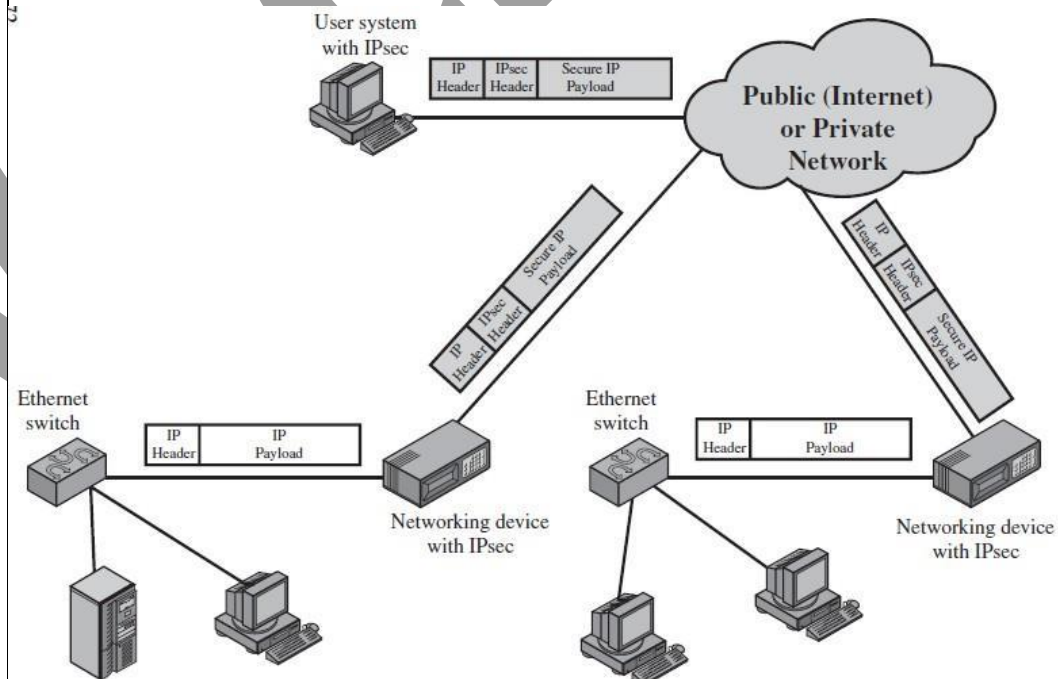
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Q. 09 a Explain benefits and routing applications of IPSec

5

6

IPsec, or Internet Protocol Security, offers several significant benefits and plays a crucial role in routing applications. Here's a detailed overview:



An IP Security Scenario

Benefits of IPsec:

1. **Comprehensive Security** : When implemented in a firewall or router, IPsec provides robust security for all traffic crossing the network perimeter. This means that any data entering or leaving the organization is protected, ensuring confidentiality and integrity.
2. **Efficiency** : IPsec operates below the transport layer (TCP, UDP), making it transparent to applications. This means that existing software on user or server systems does not need to be modified when IPsec is implemented, reducing the overhead of security-related processing within the internal network.
3. **Resistance to Bypass** : By ensuring that all traffic from the outside must pass through the firewall, IPsec minimizes the risk of unauthorized access. This is particularly important for organizations that need to maintain strict security protocols.
4. **User Transparency** : End users do not need to be trained on security mechanisms, nor do they need to manage keying material. This simplifies the user experience and reduces administrative overhead, as there's no need to revoke or issue keys when users leave the organization.
5. **Support for Various Applications** : IPsec can secure a wide range of applications, including remote logon, email, file transfer, and web access, by encrypting and authenticating all traffic at the IP level.

Routing Applications of IPsec:

1. **Secure Router Advertisements** : IPsec ensures that router advertisements come from authorized routers, preventing unauthorized devices from disrupting network communications.
2. **Neighbor Advertisements** : It verifies that neighbor advertisements, which are essential for maintaining relationships between routers in different routing domains, originate from legitimate sources.
3. **Routing Updates** : IPsec protects against forged routing updates, ensuring that the routing information exchanged between routers is authentic and reliable.
4. **Secure Branch Office Connectivity** : Organizations can create secure virtual private networks (VPNs) over the Internet, allowing branch offices to connect securely without relying solely on private networks, which can be costly.

5. Remote Access Security : Employees can securely access company networks from remote locations, reducing costs associated with toll charges for traveling employees and telecommuters.

6. Extranet and Intranet Connectivity : IPsec can secure communications with partners and other organizations, ensuring that sensitive data is transmitted securely and that authentication and confidentiality are maintained.

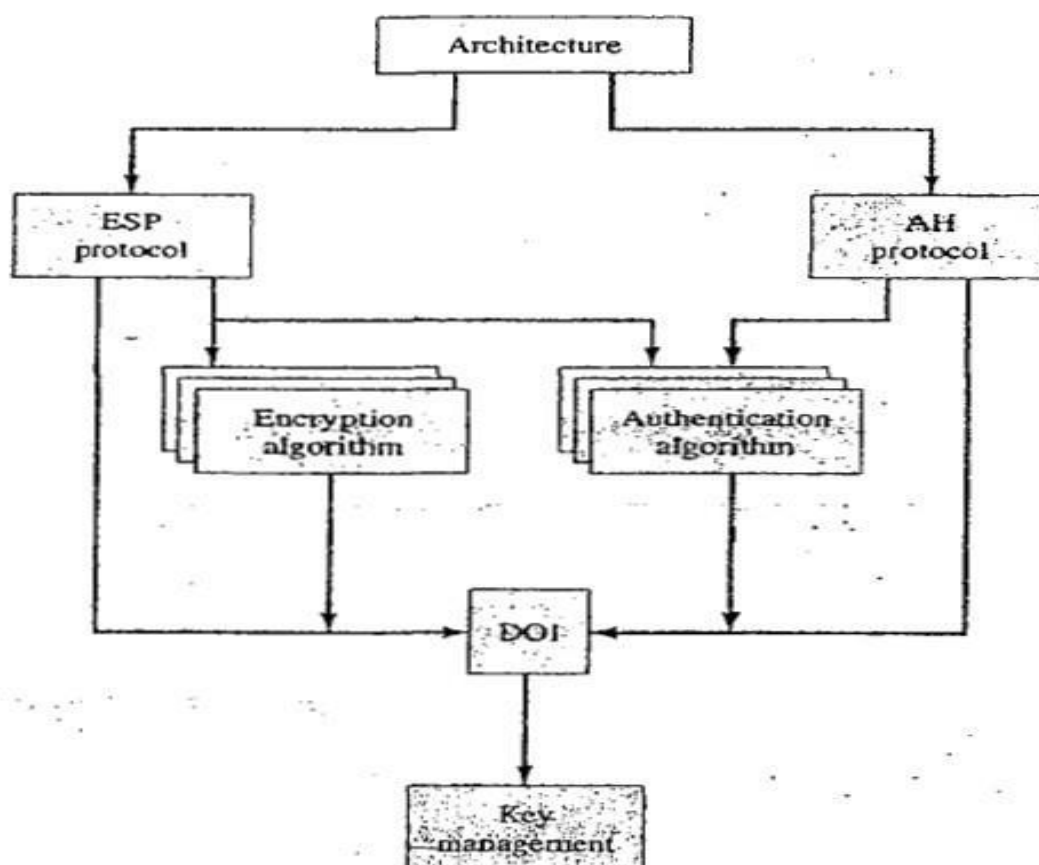
IPsec not only enhances the security of network communications but also plays a vital role in ensuring the integrity and authenticity of routing protocols, making it an essential component of modern network architecture.

b Explain IP security Architecture.

5

6

IP security architecture, commonly referred to as IPsec, is a framework designed to secure Internet Protocol (IP) communications through a suite of protocols that provide various security services. The architecture encompasses several key components and functionalities aimed at ensuring the confidentiality, integrity, and authenticity of data transmitted over IP networks.



IPSec Document Overview

1. Core Components :

- Authentication : This ensures that the data comes from a legitimate source. IPsec uses mechanisms like the Authentication Header (AH) to provide message authentication.
- Confidentiality : This is achieved through encryption, primarily using the Encapsulating Security Payload (ESP), which encrypts the data being transmitted to prevent unauthorized access.
- Key Management : IPsec includes protocols for managing cryptographic keys, which are essential for establishing secure communications. The Internet Key Exchange (IKE) protocol is commonly used for this purpose.

2. Architecture Overview :

- The IPsec architecture is defined in several RFCs, with RFC 4301 outlining the security architecture for the Internet Protocol. It specifies how IPsec can be implemented to secure both IPv4 and IPv6 traffic.
- IPsec operates at the network layer, meaning it can secure all traffic that passes through the IP layer, including various applications like email, file transfers, and web access.

3. Modes of Operation :

- Transport Mode : In this mode, only the payload of the IP packet is encrypted and/or authenticated. This is typically used for end-to-end communication between two hosts.
- Tunnel Mode : This mode encapsulates the entire original IP packet within a new IP packet, providing a secure tunnel for the data. This is commonly used for Virtual Private Networks (VPNs), allowing secure communication between different networks over the Internet.

4. Applications of IPsec :

- IPsec can be used for secure branch office connectivity, remote access for telecommuters, and enhancing the security of electronic commerce transactions. It allows organizations to create secure virtual private networks over the Internet, reducing reliance on private networks and associated costs.

5. Security Associations (SAs) :

- IPsec uses Security Associations to define the parameters for the secure communication between two parties. These associations can be established manually or automatically, depending on the network environment.

IPsec provides a comprehensive framework for securing IP communications through a combination of authentication, encryption,

	and key management, making it a vital component of modern network security strategies.		
c	Explain Security Associations with SA Parameters Security Associations (SAs) are a fundamental concept in IPsec, providing a one-way logical connection between a sender and a receiver that ensures security services for the traffic transmitted over it. If two-way secure communication is required, two separate SAs are needed. Each SA is uniquely identified by three key parameters: 1. Security Parameters Index (SPI) : This is a 32-bit unsigned integer that is assigned to the SA and is significant only locally. The SPI is included in the Authentication Header (AH) and Encapsulating Security Payload (ESP) headers, allowing the receiving system to identify which SA to use for processing a received packet. 2. IP Destination Address : This parameter specifies the address of the destination endpoint for the SA. This could be an end-user system or a network device such as a firewall or router. 3. Security Protocol Identifier : This field, found in the outer IP header, indicates whether the SA is an AH or ESP security association. In any IP packet, the combination of the destination address in the IPv4 or IPv6 header and the SPI in the extension header (AH or ESP) uniquely identifies the security association. Additionally, SAs can be organized into bundles to provide a desired set of IPsec services. There are two primary methods for combining SAs: - Transport adjacency : This method applies multiple security protocols to the same IP packet without using tunneling. It allows for a single level of combination, as further nesting does not yield additional benefits. - Iterated tunneling : This approach involves applying multiple layers of security protocols through IP tunneling, allowing for multiple levels of security to be implemented. SAs are crucial for establishing secure communication channels in IPsec, enabling various configurations and flexibility in how security services are applied to IP traffic.	5	8

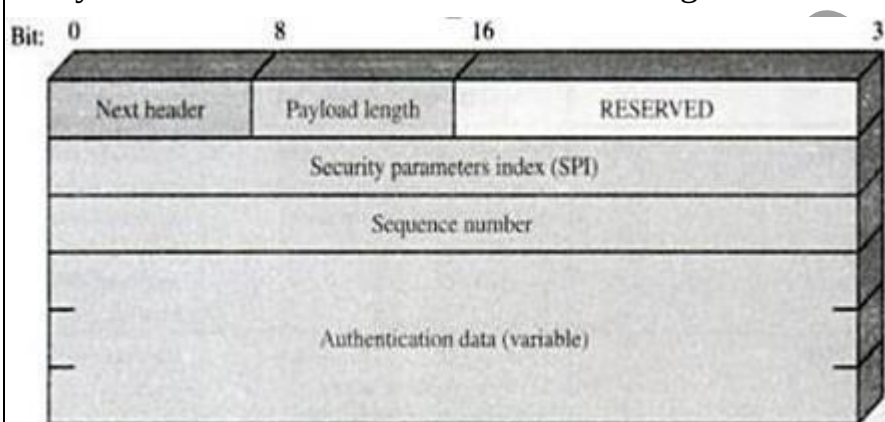
OR

Q. 10 a Explain Authentication Header with figure

5

5

The Authentication Header (AH) is an extension header used in the IPsec protocol suite to provide message authentication. Its primary purpose is to ensure the integrity and authenticity of the data being transmitted over the network. AH achieves this by adding a header to the IP packet that contains authentication information, which is used to verify that the data has not been altered during transmission.



IPSec Authentication Header

In the context of IPsec, the AH is designed to protect the entire IP packet, including the IP header and the payload, except for mutable fields in the IP header (like the Time to Live field). This means that the AH provides a way to authenticate the sender and ensure that the message has not been tampered with.

Key Features of the Authentication Header:

1. Integrity : Ensures that the data has not been modified in transit.
2. Authentication : Confirms the identity of the sender.
3. Replay Protection : Prevents attackers from capturing and re-sending packets.

Figure Explanation:

While I can't display a figure directly, I can describe what such a figure would typically illustrate:

- Original IP Packet : The figure would show an original IP packet with its header and payload.
- AH Header : The AH header would be added after the original IP header, indicating that it is an extension to the IP packet.
- Authentication Data : The figure would depict the authentication data field that is appended to the packet, which contains the hash of the packet data, ensuring integrity and authenticity.

Operation:

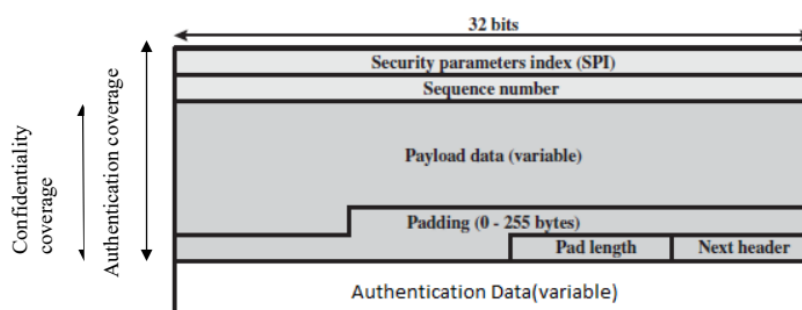
When a sender transmits a packet using AH, the following steps occur:

1. The sender creates the original IP packet.
2. A hash of the packet (including the IP header and payload) is generated.
3. This hash is then included in the AH header.
4. The entire packet, now with the AH header, is sent to the receiver.
5. Upon receiving the packet, the receiver can compute the hash of the received packet and compare it with the hash in the AH header to verify integrity and authenticity.

the Authentication Header is a crucial component of IPsec that provides a robust mechanism for ensuring the integrity and authenticity of IP packets, making it an essential part of secure communications over the Internet.

b Explain ESP header with figure

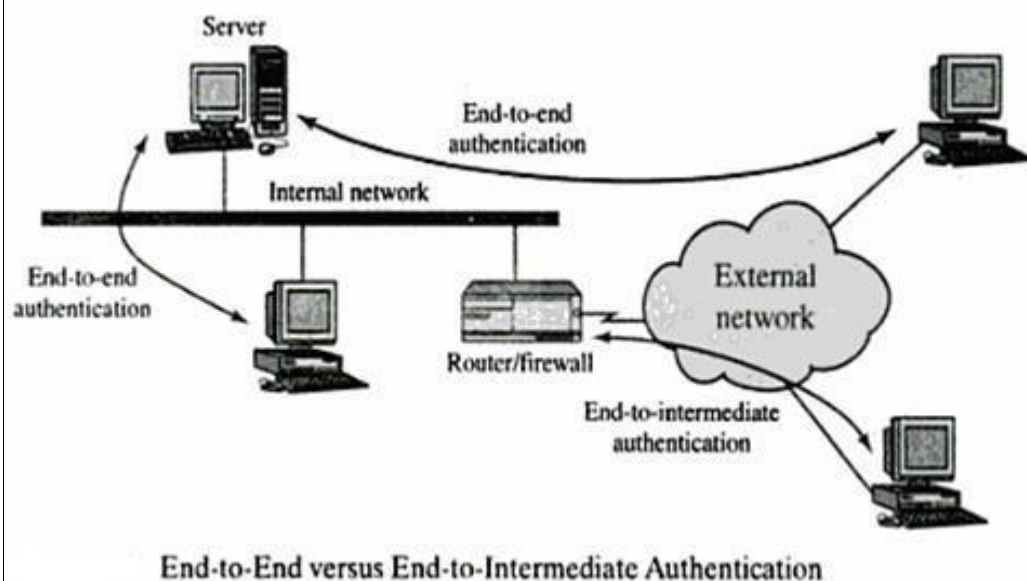
The Encapsulating Security Payload (ESP) header is a crucial component of the IPsec protocol suite, designed to provide various security services such as confidentiality, data origin authentication, and integrity for IP packets. The ESP header is structured to facilitate these services effectively.

**ESP Header Structure**

The ESP packet format consists of several key fields:

1. **Security Parameters Index (SPI)** : This 32-bit field identifies the security association (SA) in use. It helps the receiving system determine which keys and algorithms to use for decryption and authentication.
2. **Sequence Number** : Another 32-bit field that contains a monotonically increasing counter value. This sequence number is essential for providing an anti-replay function, ensuring that old packets cannot be resent by an attacker.

	3. Payload Data : This variable-length field contains the actual data being transmitted, which could be a transport-level segment (like TCP or UDP). 4. ESP Trailer : This includes padding, pad length, and next header fields. The padding is used to ensure that the payload aligns with the required block size of the encryption algorithm. 5. ESP Authentication Data : If authentication is selected, this field is added after the ESP trailer and contains the authentication data for the entire packet, ensuring that the data has not been tampered with. Modes of Operation ESP can operate in two modes: - Transport Mode : In this mode, only the payload of the IP packet is encrypted and optionally authenticated. The ESP header is inserted directly before the transport-layer header, and the entire transport segment is encrypted. - Tunnel Mode : This mode encapsulates the entire original IP packet within a new IP packet. The ESP header is added to the outer packet, allowing for secure communication between networks over potentially insecure channels, such as the Internet. The ESP header plays a vital role in securing IP communications by providing encryption and authentication services. Its structure allows for flexibility in how data is protected, whether in transport or tunnel mode, ensuring that sensitive information remains confidential and intact during transmission.		
c	Explain transport mode & Tunnel Mode of IPsec Transport mode and tunnel mode are two operational modes of the IPsec (Internet Protocol Security) protocol, each serving different purposes in securing network communications.	5	10



Transport Mode

In transport mode, IPsec provides confidentiality and optional authentication for the payload of an IP packet. This means that only the data being transmitted (the payload) is encrypted, while the IP header remains intact and visible. This mode is typically used for end-to-end communication between two hosts, such as a client and a server or between two workstations.

Key Features of Transport Mode:

- **Encryption of Payload:** The data following the IP header is encrypted, which protects the content of the communication.
- **Traffic Analysis Vulnerability:** Since the IP header is not encrypted, it is possible for an attacker to perform traffic analysis on the transmitted packets, potentially inferring information about the communication.
- **Use Cases:** Commonly used in scenarios where both communicating hosts support IPsec, allowing for direct secure communication.

Tunnel Mode

Tunnel mode, on the other hand, encapsulates the entire original IP packet (including the IP header) within a new IP packet. This mode is particularly useful for creating Virtual Private Networks (VPNs) and securing communications between networks over the Internet.

Key Features of Tunnel Mode:

- **Full Packet Encryption:** The entire inner IP packet, including the original IP header, is encrypted. This prevents intermediate routers from examining the inner packet, thus enhancing security.
- **Encapsulation:** The original packet is encapsulated with a new IP header, which allows for routing while keeping the inner packet secure. This means that the outer packet can have different source and

	destination addresses, adding an additional layer of security. - Use Cases: Ideal for scenarios where one or both ends of the communication involve a security gateway, such as a firewall or router. This allows multiple hosts behind a firewall to communicate securely without needing to implement IPsec individually. transport mode is suitable for direct host-to-host communication where only the payload needs protection, while tunnel mode is designed for securing communications across potentially insecure networks, encapsulating the entire packet to provide comprehensive security. Each mode has its specific applications and advantages depending on the network architecture and security requirements.		
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REVISED BLOOMSTAXONOMYLEARNINGLEVEL(RBT)

L1:Remember	L2:Understand	L3:Apply	L4:Analyze	L5:Evaluate	L6:Create
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COURSE OUTCOMES(COs)

1	Understand Cryptography, Network Security theories, algorithms and systems
2	Apply different Cryptography and Network Security operations on different applications
3	Analyze different methods for authentication and access control
4	Evaluate Public and Private key, Key management, distribution and certification
5	Design necessary techniques to build protection mechanisms to secure computer networks

PROGRAM OUTCOMES(POs)

1	Engineering Knowledge	5	Modern tool usage	9	Individual and Team-Work
2	Problem Analysis	6	Engineer and Society	10	Communication
3	Design / Development Solutions	7	Environment and Sustainability	11	Project Management and Finance
4	Conduct Investigations of Complex	8	Ethics	12	Life-long Learning

	problems			
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